University of Minnesota custom Azure course

Session 4 Participant Workbook

Resilience, Disaster Recovery, and Multi-Region Design

Azure Storage, Data Protection, and Resilience at Scale

Session: Day 2, Session 4, four hours

Use: Checkpoint discussion, design exercise or lab debrief, discussion prompts, knowledge check

Keep: Your worksheets are the input to the Session 6 roadmap capstone

All scenarios use synthetic data modeled on a large public research university. No real University of Minnesota data is used.

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1. Session map

Day 2, Session 4: Resilience, Disaster Recovery, and Multi-Region Design. Four hours including a fifteen-minute break.

Block	Minutes	Activity
Block 1	55	Redundancy options, object replication, and cross-region copy
Block 2	50	Azure Site Recovery and the regional loss demonstration
Break	15	
Block 3	40	RPO and RTO design method and Resiliency in Azure
Block 4	80	Design exercise and facilitated review

1.1 What this session covers

Enterprise data must be highly available or the university cannot function; research data may tolerate downtime but must never be lost; and holding multiple copies of petabytes is expensive. This session gives the vocabulary and the method: availability, durability, and recoverability as three different promises; what LRS, ZRS, GRS, GZRS, and their read-access variants guarantee and what they do not; customer-managed planned and unplanned failover; object replication and copies beyond Azure; Azure Site Recovery with recovery plans and test failover; a five-step RPO and RTO design method; and Resiliency in Azure as the single view of protection.

The design exercise builds the university resilience matrix: six representative workloads, each with redundancy, replication, backup, and DR approach, RPO and RTO, relative cost, and the accepted risk written down.

1.2 Have ready

- Your current recovery expectations for the enterprise database, the identity platform, and the learning management system, even if they are informal.
- The Session 1 workbook: whether the research data classes were placed in flat or hierarchical namespace accounts, because it changes the second-copy approach.
- The cost model workbook open if you have a laptop; the redundancy multipliers are on the Assumptions sheet.

1.3 Materials you will receive

- Session 4 deck; demo script `S04_Demo_Script_Regional_Loss_Survived`
- Redundancy option comparison reference card and the sample recovery plan definition (Reference Card Pack and the Bicep bundle)

- `UMN_Storage_Cost_Model_Workbook.xlsx` (Scenarios and EgressAndCopies sheets) for the cost lines
- This workbook (participants, including the resilience matrix worksheet) and the facilitator guide (instructor, with the model matrix)

2. Block 1 checkpoint: sort your data classes

Three questions about the university's data classes. Name classes, not systems.

Question 1

Which data classes need to survive a regional loss?

Question 2

Which need to stay available during a regional loss?

Question 3

Which need a copy outside Azure entirely?

3. Design exercise: the university resilience matrix

In groups, build the resilience matrix for six representative workloads: the enterprise database (Oracle ERP), the identity platform, learning management data, an active research dataset, a cold research archive, and a departmental file share. For each, select redundancy, replication, backup, and DR approach; assign RPO and RTO; estimate relative cost (low, medium, high, with the multiplier from the cost model if you can); and write down the accepted risk and who accepts it.

Two workloads have a twist. The active research dataset's answer depends on whether your Session 1 design put it in a flat or hierarchical namespace account. The cold research archive's second copy must be made while the data is online (or with a tool that rehydrates first), because object replication cannot read archived blobs.

3.1 Steps

1. Agree the four resilience tiers your group will use (mission critical, business important, research active, research archive) and the RPO and RTO you attach to each. Record them in Worksheet 1.
2. Place each of the six workloads in a tier and complete Worksheet 2: redundancy, replication, backup, DR, RPO, RTO, relative cost.
3. Complete Worksheet 3: the accepted risk for each workload and who accepts it (a role, not a person).
4. For the two workloads with twists, write one sentence each explaining the second-copy mechanism.
5. Prepare to defend the matrix in the facilitated review against the four review lenses.

3.2 Worksheet: resilience tiers

Tier	Typical RPO	Typical RTO	Azure building blocks	Relative cost
Mission critical				
Business important				

Tier	Typical RPO	Typical RTO	Azure building blocks	Relative cost
Research active				
Research archive				

3.3 Worksheet: the six workloads

Workload	Tier	Redundancy	Replication or second copy	Backup	DR approach	RPO	RTO	Cost
Enterprise database (Oracle ERP)								
Identity platform								
Learning management data								

Workload	Tier	Redundancy	Replication or second copy	Backup	DR approach	RPO	RTO	Cost
Active research dataset								
Cold research archive								
Departmental file share								

3.4 Worksheet: accepted risk

Workload	What is NOT protected against (the accepted risk)	Who accepts it (role)
Enterprise database		
Identity platform		
Learning management data		

Workload	What is NOT protected against (the accepted risk)	Who accepts it (role)
Active research dataset		
Cold research archive		
Departmental file share		

3.5 What you present

- The completed matrix (Worksheet 2) and the accepted-risk table (Worksheet 3).
- Two sentences: the second-copy mechanism for the active dataset and for the cold archive.
- The one workload where the group disagreed, and how it was settled.

3.6 How your design will be critiqued

Lens	Meets	Exceeds
RPO and RTO match business impact	Tiers defined with numbers; each workload placed with a reason	Different RPO and RTO for data loss versus availability on the same workload
Redundancy covers the feared failure	Zone versus region loss named per workload	Account failover and feature conflicts (object replication, NFS) acknowledged
Recoverable copy outside the blast radius	Backup or replication separate from the primary administrative scope	A copy outside Azure for the archive, with integrity manifests
Honest cost line and written risk	Relative cost and accepted risk stated for every workload	Multipliers from the cost model used; risk owner is a named role

4. Discussion prompts

Work through these in your group or with a neighbor during the time the facilitator calls out. Write enough that you can defend the position in the room.

Prompt 1

Availability, durability, recoverability: for the identity platform, which one does the university have today, and which one is it missing?

Block 1 opening.

Prompt 2

The planned account failover in the demo took time and made the account unavailable for writes during the swap. How would you schedule a DR drill around that for the enterprise application?

Block 2.

Prompt 3

Resiliency in Azure shows protection status for every vault and replicated item. What would you put on a monthly review agenda from it?

Block 3.

5. Knowledge check

Six questions, one best answer each. Mark your answer; the facilitator reviews them at the end of the session.

1. Which redundancy option keeps data readable and writable if one availability zone is lost?

- A. LRS
- B. ZRS
- C. GRS
- D. RA-GRS

2. What happens to geo-redundancy after an unplanned customer-managed failover?

- A. It is retained
- B. The account becomes locally redundant in the new primary until geo-redundancy is re-enabled
- C. The account is deleted
- D. The secondary becomes read-only forever

3. Which blob type can object replication copy?

- A. Block blobs
- B. Append blobs
- C. Page blobs
- D. Snapshots

4. What does a Site Recovery test failover do to production replication?

- A. Stops it
- B. Reverses it
- C. Nothing; test failover runs in an isolated network
- D. Deletes the recovery points

5. Which recovery point type gives the lowest RTO?

- A. Latest

- B. Latest processed
- C. Latest app-consistent
- D. Latest multi-VM processed

6. How many protected instances can one recovery plan include?

- A. 10
- B. 50
- C. 100
- D. Unlimited

6. Takeaways and the capstone

- Redundancy protects against infrastructure failure, not deletion or corruption; availability, durability, and recoverability are different promises.
- Object replication and copies outside Azure add independence at a known cost; replicate archives while they are online.
- Site Recovery plans and test failovers turn DR from hope into rehearsal; planned account failover keeps geo-redundancy, unplanned does not.
- The resilience matrix is a business document: RPO and RTO per class, cost per tier, and the accepted risk with its owner.

Capstone input from this session

University resilience matrix applied to six workload classes.

Next: Day 3 opens with Session 5, next-generation protection for Oracle and enterprise databases: the same principles applied to the university's largest databases.

7. Notes

Your notes

